



STERLING AND WILSON PVT. LTD.

MATERIAL INDENT FORM

(INTERNATIONAL OPERATION DIVISION)

INDENT NO.::

SW-HYD02-HVAC-16

REV :

00

JOB NAME :

HYD02 - HYDERABAD DATA CENTER

DATE.:

25/11/2024

REV DATE.:

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JOB NO.:

NOT ASSIGNED

ITEM UNDER TYPE OF CONTRACT :

S.N	ITEM / MATERIAL DESCRIPTION	QTY	UNIT	BOQ REF. No.	OWN COST (INR)				REMARKS
					MATERIAL		LABOUR		
					UNIT RATE	TOTAL AMOUNT	UNIT RATE	TOTAL AMOUNT	
	CHILLED WATER PIPE INSULATION								Refer Technical Specification: HYD02-M-S-008-0-SECTION 23 07 00 APAC-HVAC INSULATION.
1.0	Supply, installation, testing and commissioning of Chemically blown closed cell nitrile rubber in tubular sections and sheet for pipe insulation, Comply with ASTM C 534, Type I on chilled water pipes and fittings like valves, flanges and unions,Adhesive, etc. as per approved shop drawings and specifications. Density of the nitrile rubber shall be 40-60 Kg/m3.he product shall have temperature range of -40 oC to 105oC. Tubular Section upto Pipe size: Dai. 100 mm. Nitrile rubber sheet: Above Dia. 100 mm. Insulation thickness as per below details: Pipe size: Dia. 20 mm - 150 mm- 32 mm. thk. Pipe size: Dia. 200 mm - Dia. 450 mm - 50 mm thk. I.e. 2 layers of 25 mm each. Pipe size: Above Dia. 500 - 75 mm thk.I.e. 3 layers of 25 mm each. Refer technical specifications Section : HYD02-M-S-008-0 23 07 00 APAC form details.								
a)	Insulation works for Piping of ϕ 900 mm	394.8	Rmt			₹ -	₹ -	₹ -	
b)	Insulation works for Piping of ϕ 450 mm	378	Rmt			₹ -	₹ -	₹ -	
c)	Insulation works for Piping of ϕ 400 mm	100.8	Rmt			₹ -	₹ -	₹ -	
d)	Insulation works for Piping of ϕ 350 mm	100.8	Rmt			₹ -	₹ -	₹ -	
e)	Insulation works for Piping of ϕ 300 mm	100.8	Rmt			₹ -	₹ -	₹ -	
f)	Insulation works for Piping of ϕ 200 mm	2636	Rmt			₹ -	₹ -	₹ -	
g)	Insulation works for Piping of ϕ 100 mm	2150	Rmt			₹ -	₹ -	₹ -	
h)	Insulation works for Piping of ϕ 80 mm	5	Rmt			₹ -	₹ -	₹ -	
i)	Insulation works for Piping of ϕ 50 mm	146	Rmt			₹ -	₹ -	₹ -	
j)	Insulation works for Piping of ϕ 40 mm	10.5	Rmt			₹ -	₹ -	₹ -	
k)	Insulation works for Piping of ϕ 25 mm	125	Rmt			₹ -	₹ -	₹ -	
l)	Insulation works for Piping of ϕ 20 mm	132.5	Rmt			₹ -	₹ -	₹ -	
2	CONDENSATE DRAIN WATER PIPE INSULATION								
	Supply, installation, testing and commissioning of Chemically blown closed cell nitrile rubber in tubular sections and sheet for pipe insulation, Comply with ASTM C 534, Type I on chilled water pipes and fittings like valves, flanges and unions,Adhesive, etc. as per approved shop drawings and specifications. Tubular Section upto Pipe size: Dai. 100 mm. Insulation thickness as per below details: Pipe size: Dia. 20 mm - 150 mm- 25 mm. thk. Refer technical specifications Section 23 07 00 APAC form details.								
2.1	Insulation works for Piping of ϕ 32 mm	786.5	RM			₹ -	₹ -	₹ -	
	Protection: Metal sheath insulation to Metal sheathing where: a. Exposed to sunlight. b. Subject to mechanical damage. Alternative protection: Where exposed to sunlight but not exposed to mechanical damage, provide 2 coats of tintable, water-based, rubberised, UV resistant, flexible paint finish to outdoor installations Metal sheathing 1. Material: As documented and as follows: 2. Metallic-coated sheet steel, 0.55 mm minimum thickness coating class 2275. 3. Aluminium sheet, 0.55 mm thick								
	ADHESIVE: Adhesives shall comply with maximum VOC limits specified in Section 01 81 13.14, Sustainable Design Requirements – LEED v4 BD+C. B. Materials shall be compatible with insulation materials, jackets, and substrates and for bonding insulation to itself and to surfaces to be insulated, unless otherwise indicated.								
	TAPES: 1. ASJ Tape: White vapor-retarder tape matching factory-applied jacket with acrylic adhesive, complying with ASTM C 1136. 2. Width: 3 inches/76mm. 3. Maximum thickness: 0.14mm. 4. Adhesion: 90 ounces force/inch in width. 5. Elongation: 2 percent. 6. Tensile Strength: 490 N/100mm 7. ASJ Tape Disks and Squares: Precut disks or squares of ASJ tape.								

COMPLIED FOR INSULATION MATERIAL



S.N	ITEM / MATERIAL DESCRIPTION	QTY	UNIT	BOQ REF. No.	OWN COST (INR)				REMARKS
					MATERIAL		LABOUR		
					UNIT RATE	TOTAL AMOUNT	UNIT RATE	TOTAL AMOUNT	
	PRODUCT DELIVERY AND STORAGE A. Deliver material properly labeled, packaged and undamaged. B. Do not store exposed to weather, provide suitable material to protect from damage.								
	2.1 INSULATION 2.1.1 General Insulation Requirements: 1. Insulation shall be applied on clean dry surfaces, after pressure testing inspection and release for insulation application. 2. Insulation shall be continuous through wall, floor, ceiling openings and sleeves. Where insulated piping or ductwork pierces fire rated partitions, walls and floors, substitute anhydrous calcium silicate insulation with vapour barrier. 3. Install multiple layers of insulation with longitudinal and end seams staggered. 4. Install insulation with tight longitudinal seams and end joints. Bond seams and joints with adhesive recommended by insulation material manufacturer. 5. Where vapor barrier is indicated, seal joints, seams, and penetrations in insulation at hangers, supports, anchors, and other projections with vapor-barrier mastic. 6. Where vapor barrier is indicated, seal joints, seams, and penetrations in insulation at hangers, supports, anchors, and other projections with vapor-barrier mastic.								
REQUIREMENT / THINGS TO BE PROVIDED BY MANUFACTURE / VENDOR :					SPECIAL NOTES IF ANY				
1. TECHNICAL SPECIFICATION SECTION TO BE COMPLIED WITH SIGN & STAMP BY VENDOR.		4. COMMISSIONING AND MANDATORY SPARE PART LIST							
2. a) TECHNICAL DATA SHEET TO BE FILLED BY VENDORS. b) Vendor Required to Submit manufacturer's product data, including printed technical literature, installation instructions and catalogue.		5. QUALITY ASSURANCE PLAN / INSPECTION TEST PLAN							
3. TECHNICAL DEVIATION LIST IF ANY WITH RESPECT TO TENDER SPECIFICATION		6. G.A DRAWING / FOUNDATION DWG / PRODUCT CATALOGUE							

1. TRANSPORTATION :	Included in above rates	LIST OF APPROVED MAKE		
2. GST :	As Applicable			
3. PORT CLEARANCE :	NA			
4: VAT @ 14 % :	NA	DELIVERY REQUIRED BY :		
5 : LD / PENALTY :	NA	LIST OF ATTACHMENTS		
6. IMPORT DUTIES	NA	1. TECHNICAL SPECIFICATION FOR HVAC INSULATION		
7 : DUTY BENEFIT IF ANY FOR IMPORTED ITEMS	NIL			
8. WARRANTY / D L P PERIOD	24 MONTHS FROM DATE OF DELIVERY OR 18 MONTHS FROM DATE OF COMMISSIONING			

DEPARTMENT :	PREPARED BY :	CHECKED BY:	AUTHORIZED BY :	SHEET :
DESIGN	NIKITA SANGALE	SADIQ SHAIKH		1 of 1